

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461077.0574 +/- 0.007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.028 +/- 0.052
Transit depth (Rp/Rs)^2	0.081 +/- 0.29 [%]
Orbital Inclination (inc)	86.13 +/- 1.8
Airmass coefficient 1 (a1)	0.002651 +/- 1.8e-05
Airmass coefficient 2 (a2)	3.0 +/- 1.34
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#3 - [621, 172]
Optimal Aperture	27.16
Optimal Annulus	56.53
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.028 $\pm$ 0.052 vs 0.1326 (deviation 2.01 σ)
Duration fit vs published	0.0 d vs 0.1238 d (deviation inf σ)
Mid-time O-C	-5669.0 min
Depth SNR	0.5
Residual scatter	1.15 %

Light curve

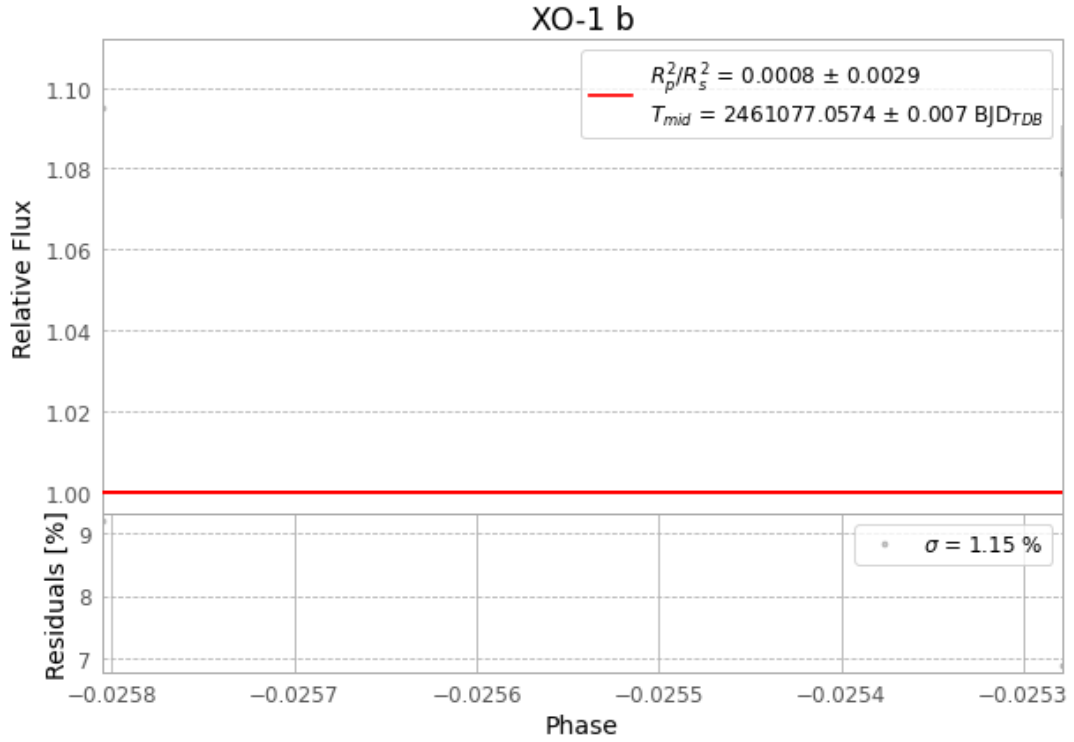


Figure 1 — FinalLightCurve_XO-1 b_2026-02-09.png (SHA-256 81fca0f697144586...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 134], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9fb1e4dfe665020a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

77 FITS frames (51 MB), XO-1260209092715.FITS → XO-1260209131515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-260209.log (1b8340e495a2...), FinalLightCurve_XO-1 b_2026-02-09.png (81fca0f69714...), FinalLightCurve_XO-1 b_2026-02-09.csv (7dd073b47c37...)

Compute resources

Wall time 219.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.